Wealth Management Americas IT

Key Elements of an Architecture Strategy

Discussion with WMA COO - 8 October 2010

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Executive Summary

- Target-State Architecture (TSA) was designed to be shared across WMA business as input into 2011 planning process
- Approach -- Front to back Target-State architecture mind-set
  - Technical deep dive sessions over 8 weeks with over 60 participants (see Attachment A)
  - Elements of the Design for Six Sigma methodology toolkit (e.g. QFD, see Attachment B)
  - Twenty key principles identified and ranked to derive the fifteen architecture tenets of the candidate TSA
  - Provides pro-forma implementation plan for activities in the next years and a way of tracing the investment initiatives under-review to the tenets of the TSA
- Proposed TSA is
  - Aligned with existing strategies formulated by WMA business, by the Global CTO group and GTEC as a whole
  - Articulated in a way relevant to the business for meaningful interaction between business and IT
  - A framework to make standards (logical architecture) and implement approach (physical architecture) decisions
  - The foundation of an actionable agenda: 40+ activities with relevance directly traceable back to key principles
  - The possible underpinning of the balance scorecard/KPIs to measure progress
- In 2011, we will focus on enhancing the platforms that support revenue increase (e.g. Banking, AI, etc) along with strategic core architecture changes, e.g.:
  - Consolidation of data stores onto golden sources
  - Re-design front systems with decoupled presentation layer from business logic
  - Deployment of key components of messaging and business infrastructure as we re-platform and refresh the application portfolio front to back
- Next steps:
  - Validation in principle by the business and GTEC executive management (tenets, portfolio of initiatives)
  - Prepare and contribute to the workshops set up to develop WMA front-to-back strategy and investment plans (in progress)
  - Set up the process and organizational infrastructure to support and monitor the execution of the strategy
  - Additional consideration for other aspects/views of the target state needed (e.g. security and infrastructure)

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A – Participants

B - Using Design for Six Sigma tools for Architecture Strategy Definition

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Background and Objectives

Following the CTO Assessment of the WMA architecture, WMA IT has developed a target state application & data architecture to be shared with the business as input into the 2011 planning process.

Objectives and Scope

- Develop front-to-back conceptual and logical architectural views for :
  - Functional, Application and Data for WMA as a whole
  - Other relevant views for select domain area (e.g. Front Office)
  - Core foundation of the investment governance process/decision making
- Articulate findings for presentation to key stakeholders: for review and validation
  - Enable the business change agenda (WMA and UBS) and aligned with business strategy
  - Implement GTEC agenda in strategic directions (in particular the Group-wide App Strategy)
  - Foundational reference for development of domain architectures usable throughout WMA

Sources

- Relied on Area Heads for:
  - Current state,
  - Business units objectives
  - Current initiative
  - Existing strategic programs in progress for their areas.
- Discussions with CTO functions within Group Technology to leverage existing assets

Method

- Agile: Issues and challenges relatively well known: little time spent on “as is” analysis
- Practical: Focus on actionable agenda with right mix of “hygiene” action items and steps that lead to true competitive advantages, not merely incremental improvements
- Disciplined: Bias towards presenting the “why” as much as the “what” with explicit description of traceability of recommendations to stated principles
- Pragmatic: By necessity, our initial step was bottom up to harvest information but recommendations and plans were conceived as an integrated whole
- Generally followed reference methodology and design/format as per existing Group CTO guidelines

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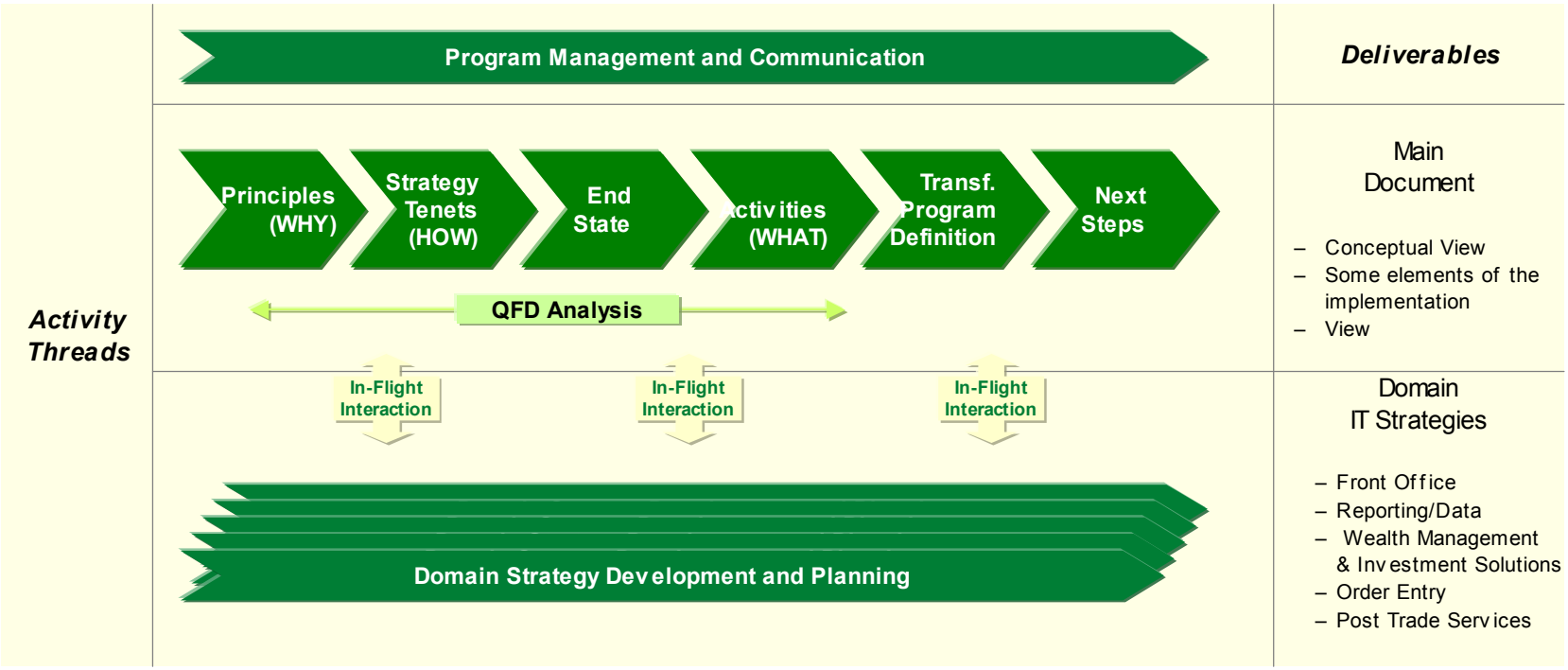
Project Summary and Methodology

- Based on GTEC Strategy and IT Principles (4Q 2009), IT Strategy (1Q 2010) and CTO Assessment / Application Health check (3/8 – 5/10)
- Multi-track / Multi-wave approach – with large group of participants (Attachment A)
  - Bottom up analysis
  - Top down synthesis
- Some elements of the Design for Six Sigma (QFD Analysis)



This presentation proposes an actionable agenda built in continuity of previous discussions and in preparation for further discussions with all stakeholders.

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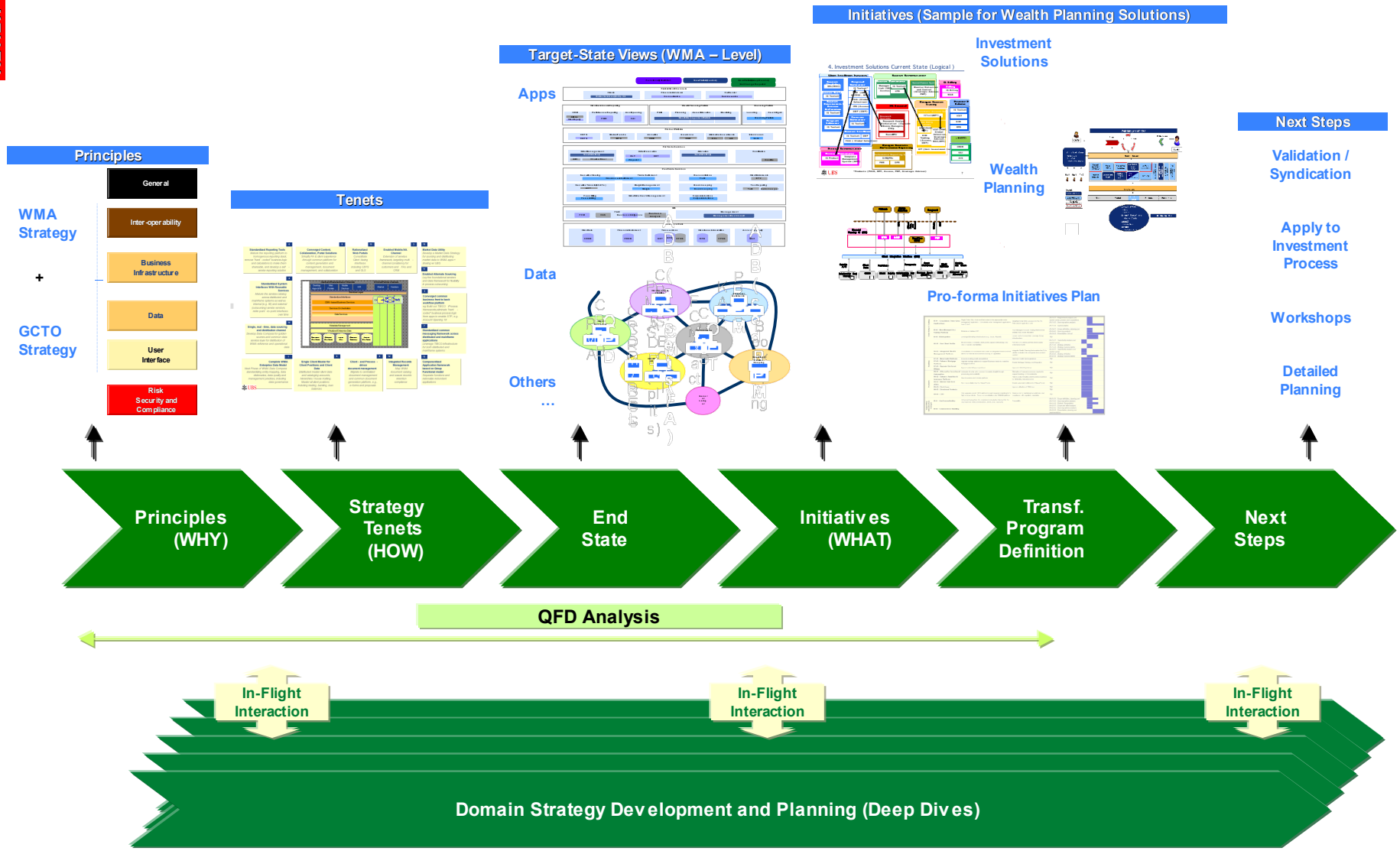
Case Study: Wealth Planning Solutions Deep Dive



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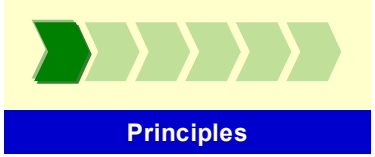
At the conceptual level, the overall target IT Architecture is based on a series of key principles in six fundamental domains…



WMA Business Strategy

- Advice-focused, not product-focused
- Anticipate Client Needs
- The best, not the biggest
- Superior Client Experience

+

Group Technology Strategy

- Improve quality of solutions
- Define differentiating Group architecture
- Simplify application and infrastructure suite
- Build strong and vibrant technical community

General	Strive for Cost-efficient Design and Development
	Support an Evolutionary Approach
	Follow the Group Functional Model (CBM)
	Design for Failover and Recoverability
	Design for Alternate Sourcing
Inter-operability	Use Standard Systems and Protocols for External Communication
	Use Standard Communication and Interaction Patterns Between Applications (Eg Event-based)
	Strive for Self-contained Loosely Coupled Applications Facilitating STP
Business Infrastructure	Re-use Software Components / Services / Solutions
	Standardize Business Infrastructure Such As Doc Mgmt. Msg, Workflow
	Provide Multi Currency Capability
	Enable Flexible and Efficient Product Configuration (Order Mgmt Framework)
	Architect Different Systems for Operational, Analytical and Archiving Purposes
Data	Define Unique Golden Sources & Assign Business and IT Ownership for Each Data Object
	Reduce Redundant Business Data Stores - Allow Controlled Data Redundancy, Under Control of Data Owner
	Follow Core Data Quality Principles (Eg Correct at Source)
User Interface	Provide User Group Specific Platforms Via Common Framework (FA, Ops, Client)
	Provide Channel-independent Solutions (Multi Channeling, Eg Web IVR, Mobile, Contact Center)
	Enable Consistent Client and FA / CSA / Operations View Sourcing the Same Data
	Avoid Business Logic in Front-end Components
Risk Security and Compliance	*Ensure Accountability by Providing Data for Regulatory / Legal Reporting*
	Provide Audit Trail in Compliance With Legal Discovery & Retention Requirements
	Protect All Assets According to Their Level of Confidentiality.
	Comply With Security Polices Regarding ID Mgmt, Authentication, Entitlements, Access Control

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… which we ranked based on our understanding of the firm's overall technology strategy and WMA business demands

This exercise should also be performed with stakeholders to facilitate a structured discussion about priorities to implement a true front to back architecture.

Description	Normalized Relative Importance Score
Define Unique Golden Sources & Assign Business and IT Ownership for each Data Object	0.59
Enable Consistent Client and FA / CSA / Operations View Sourcing the same Data	0.57
Avoid Business Logic in Front-end Components	0.56
Provide Channel-independent Solutions (Multi Channeling, e.g. Web IVR, Mobile, Contact Center)	0.56
Reduce Redundant Business Data Stores Allow Controlled Data Redundancy, under Control of Data Owner	0.55
Enable Flexible and Efficient Product Configuration (Order Mgmt Framework)	0.54
Provide User Group Specific Platforms Via Common Framework (e.g. FA, Ops, Client)	0.53
Follow Core Data Quality Principles (e.g. Correct at Source)	0.50
Use Standard Communication and Interaction Patterns between Applications (e.g. event-based)	0.46
Design for alternate sourcing	0.46
Strive for Self-contained Loosely Coupled Applications	0.45
Follow the Group Functional Model (CBM)	0.44
Provide Multi Currency Capability	0.42
Re-use Software Components / Services / Solutions	0.42
Design for Failover and Recoverability	0.42
Strive for cost-efficient design and development	0.41
Use Standard Systems and Protocols for External Communication	0.40
Standardize Business Infrastructure such as Doc Mgmt. Msg, Workflow)	0.38
Support an Evolutionary Approach	0.35
Architect different Systems for Operational, Analytical and Archiving Purposes	0.25

- *Ranking based on pair-wide comparison on scale of -9 (when one principle is much less important than another to 9 much more important) – see attachment B*
- *Risk security and compliance score not included (all 58 to 81) so as to focus on levers within control of WMA*

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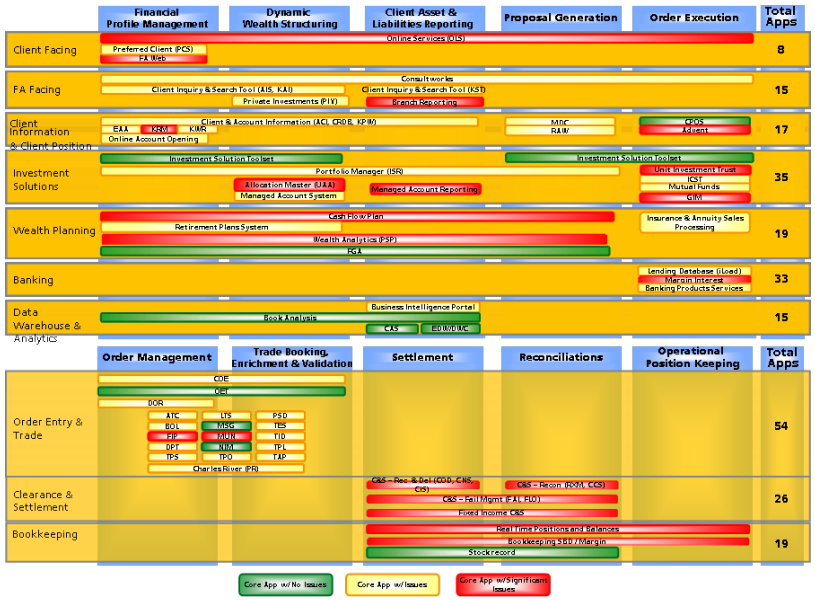
Our objective is to address the fundamental shortcomings of the current state architecture…

The current state is not fit for purpose, is overly complex and highly fragmented. This complexity impedes the business rather than enables it.



Source: Updated from 2010 CTO Review, using CBM
Note: Diagram only illustrates Core Application Domains

Current State at a Glance

- WMA is supported by 364 applications (excludes Finance and Corporate Systems)
- Of these applications, 59 are estimated to be the principal applications
- 2 applications present immediate systemic risk to the organization and IT will mobilize a "deep dive" on these (e.g. OLS, Bookkeeping)
- No overall target architecture has been adhered (with the exception of CPOS implementation) and little/no architectural governance has been put in place
- High levels of functional duplication and redundancy
- Data Warehousing & Analytics, FA Facing, Investment Solutions, Order Entry, Document Management & Archiving, Clearance & Settlement
- Significant under-investment in IT, particularly in Post-Trade Services & Banking Products
- Absence of a common approach to data sourcing and integration

Business Implications

- No single view of customer
- FA and Client frustration about how long it takes for transactions such as 24 hours for transfers and trades
- Continuous dependency on product (not client) business model
- Poor usability of system and thus reduced productivity for FAs
- Frustrating end-user experience (several day of latency in getting information propagated)
- Poor understanding of exposures
- Operational risks
- Potential for inconsistent billing
- Make barrier of entry in new business even higher
- Slow product development

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… by using key tenets (identified in the deep-dive sessions) that can directly be traced to the architectural principles

Tenets

Key Tenets	Normalized Score (*)
Complete WMA Enterprise Data Model -- Next Phase of WMA Data Compass standardizing entity mapping, data dictionaries, data quality and management practices, including data governance	0.65
Enabled Multiple Channels -- Extension of services framework, targeting multi-channel consistency for customers and FAs and CRM	0.50
Converged Common Business Workflow platform -- e.g. Build out TIBCO iProcess frameworks and eliminate "hard-coded" business process logic from applications to enable STP, e.g. Account Opening 1X	0.44
Standardized System Interfaces With Reusable Services - Mature the services catalog across distributed and mainframe systems as well as internal (e.g. IB) and external (outsourcing) service vendors; retire point-to-point interfaces over time	0.43
Single Client Master for Client Positions and Client Data -- Distributed master client data and cataloging accounts, hierarchies / house holding. Master all client positions including trading, banking, loan balances	0.41
Standardized Reporting Tools -- Mature the reporting platform to homogenous reporting stack, remove "hard-coded" business logic and calculations to make them shareable, and develop a self-service reporting solution	0.40
Standardized common messaging framework across distributed and mainframe applications -- Leverage TIBCO infrastructure for both distributed and mainframe systems	0.37
Client- and Process-driven document management -- Migrate to centralized document management and common document generation platform, e.g., e-forms and proposals	0.35
Converged Content, Collaboration, Portal Solutions -- Simplify FA experience through common platform for content generation and management, document management, and collaboration	0.35
Rationalized Web Portals -- Consolidate Client-facing interfaces including CEFS and OLS	0.34
Enabled Alternate Sourcing -- Lay the foundational services and data framework to provide flexibility in process outsourcing	0.34
Integrated Records Management Map - WMA document catalog and assure records retention compliance	0.33
Componentized Application Framework based on Group Functional Model - Separate functions and rationalize redundant applications	0.31
Single, real-time, data sourcing and distribution channel -- Develop Data Compass for golden sources and common data services layer for distribution of WMA reference and operational data	0.31
Rationalized Market Data -- Develop a Market Data Strategy for sourcing and distributing market data to WMA applications with possible sharing with other Bus	0.20

See attachment B for description of methodology used to develop ratings

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We used these principles to rate the key tenets of the Target State Technical Architecture (WMA - Level Conceptual View)

Tenets - (HOW?)

6
Standardized Reporting Tools
Mature the reporting platform to homogenous reporting stack, remove "hard-coded" business logic and calculations to make them shareable, and develop a self-service reporting solution

9
Converged Content, Collaboration, Portal Solutions
Simplify FA experience through common platform for content generation and management, document management, and collaboration

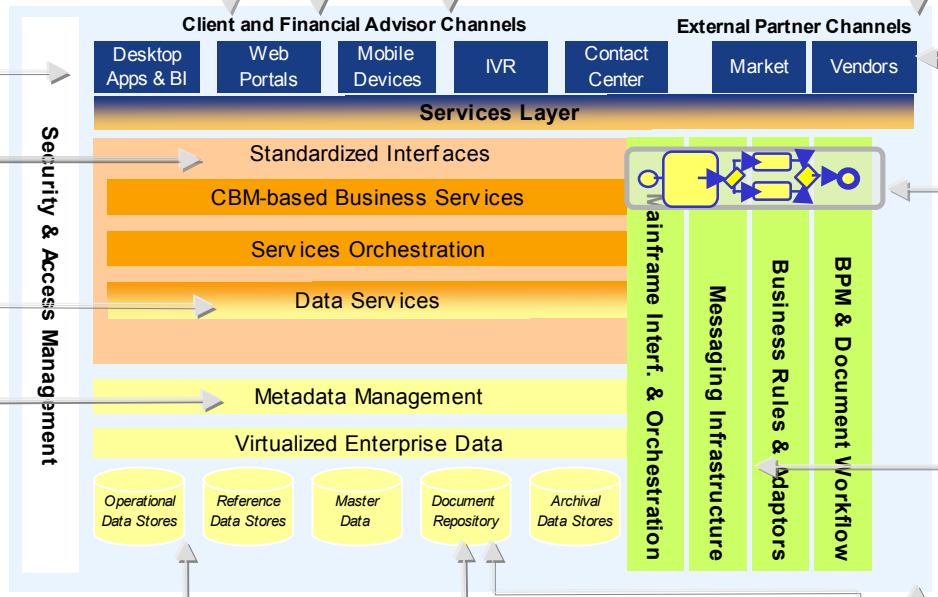
10
Rationalized Web Portals
Consolidate Client-facing interfaces including CEFS and OLS

2
Enabled Multiple Channels
Extension of services framework, targeting multi-channel consistency for customers and FAs and CRM; e.g. mobile

15
Rationalized Market Data
Develop a Market Data Strategy for sourcing and distributing market data to WMA apps + sharing w/UBS (Financial Instr.)

11
Enabled Alternate Sourcing
Lay the foundational services and data framework for flexibility in process outsourcing

4
Standardized System Interfaces With Reusable Services
Mature the services catalog across distributed and mainframe systems as well as internal (e.g. IB) and external (outsourcing) service vendors; retire point-to-point interfaces over time



3
Converged common business front to back workflow platform
e.g. Build out TIBCO iProcess frameworks, eliminate "hard-coded" business process logic from apps to enable STP, e.g. Account Opening 1X

14
Single, real-time, data sourcing and distribution channel
Develop Data Compass for golden sources and common data services layer for distribution of WMA reference and operational data

7
Standardized common messaging framework across distributed and mainframe applications
Leverage TIBCO infrastructure for both distributed and mainframe systems

1
Complete WMA Enterprise Data Model
Next Phase of WMA Data Compass standardizing entity mapping, data dictionaries, data quality and management practices, including data governance

5
Single Client Master for Client Positions and Client Data
Distributed master client data and cataloging accounts, hierarchies / house holding. Master all client positions including trading, banking, loan balances

8
Client- and Process-driven document management
Migrate to centralized document management and common document generation platform, e.g., e-forms and proposals

12
Integrated Records Management
Map WMA document catalog and assure records retention compliance

13
Componentized Application framework based on Group Functional model
Separate functions and rationalize redundant applications

UBS

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Current State Architecture - Application View

We have an architecture that requires investment front to back; we need to agree the areas where there really is a differentiating asset to 'own' or if we can increase efficiency through buying services.

Significant Issues | Some Issues | No Issues

Field & Client Channels

- Clients: Online Services (OLS)
- Financial Advisors: Consultworks
- Call Center: Serviceworks

Client Services & Reporting

- CRM: Airframe, Client Inquiry
- Performance Reporting: PMR, Advent
- Acct. Opening: CAI

Wealth Planning Platform

- Profile: IS Toolset
- Planning: FGA, NaviPlan
- Asset Allocation: MAS
- Monitoring: MAR

Banking Platform

- Lending: LON, iLoad, INT
- Cash Mgmt: Billpay

Product Platforms

- CEFS: CEFS
- Mutual Funds: MFG
- Annuities: EBIX
- Insurance: EBIX
- Alternative Investments: ECL, AIC
- Stock Loan: SLN

Pre-Trade Services

- Order Management: COE, DOR, ICST, UIT, GMI, GIM2
- Order Execution: OLT, OET, Pinpoint
- Allocation: TAPS, TUBA, Spreadsheet Macros
- Confirmation: Confirms

Post-Trade Services

- Securities Clearing: P&S, CNS
- Trade Settlement
- Reconciliations: RXM, FAL
- Client Statements: STV
- Securities Transfer (ACATs): ACT
- Margin Management: SBD
- Bookkeeping
- Tax Reporting: Tefra, Gainskeeper
- Fees / Billing: FBS, MAB
- Street Side Cash Management
- Corporate Actions: KAS, KAE, CRS

MIS

- FCM
- CAS
- Field: Business Intelligence
- Business Analysis
- Management

Reference Data

- Client Data: CRDB, REL, RMA
- Financial Instrument: PRDB, SGS, KARS, FPM
- Transactions: OET, EDW, RPB, Bkp
- Client Assets & Liabilities: CPOS, RPB, KAI, OLB, ADB
- Access / Security: MAC

CTO Assessment / Application Healthcheck(3/8)

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Target-State Architecture – Application View

… but it is critical that we effect the strategic core architecture changes we need to reach the target state (TS) which will include building or acquiring new functionalities…

Functionality Build-Out | New Platform (Custom) | New Platform (Group/Vendor) | No Change Required

Field & Client Channels

- **Clients** – Online Services (OLS) – 2.0
- **Financial Advisors** – Consultworks
- **Call Center** – Serviceworks

Client Services & Reporting

- **CRM** – Airframe; Client Inquiry
- **Performance Reporting** – PMR
- **Acct. Opening** – CAI

Wealth Planning Platform

- **Profile**, **Planning**, **Asset Allocation**, **Monitoring** – Wealth Management Platform

Banking Platform

- **Lending**, **Cash Mgmt** – Banking Platform

Product Platforms

- **CEFS** – CEFS
- **Mutual Funds** – MFG (3)
- **Annuities** – EBIX
- **Insurance** – EBIX
- **Alternative Investments** – ECL; AIC
- **Stock Loan** – SLN

Pre-Trade Services

- **Order Management** – New Order Entry; GMI (1); Managed Account Platform
- **Order Execution** – OLT; OET; Pinpoint (2)
- **Allocation** – New Order Entry
- **Confirmation** – Confirms

Post-Trade Services

- **Securities Clearing**, **Trade Settlement** – Clearance & Settlement
- **Reconciliations** – TLM
- **Client Statements** – STV
- **Securities Transfer (ACATs)** – ACT
- **Margin Management** – Margin
- **Bookkeeping** – Bookkeeping
- **Tax Reporting** – Tefra; Gainskeeper
- **Fees / Billing** – Fees & Billing
- **Street Side Cash Management**
- **Corporate Actions** – Corporate Actions

MIS

- **Field** – FCM; CAS; Business Intelligence; Business Analysis
- **Management** – Management Dashboard

Reference Data

- Products & Services
- Client Accounts
- Client Assets & Liabilities
- Client/Party
- Transactions
- Financial Instruments
- Operational Position Keeping
- General Ledger
- Access & Security

See Next Page

(1) Future and Commodity trading platform to be integrated with IB GMI.
(2) - Orders entered through WMFIX and OE platform with specific and/or algorithmic instructions will be delivered to the Retail Block Desk or Algorithmic engine.
(3) For Discussion - Capabilities move into new OET system (TBD: Move NOE and OET into Mutual Funds)

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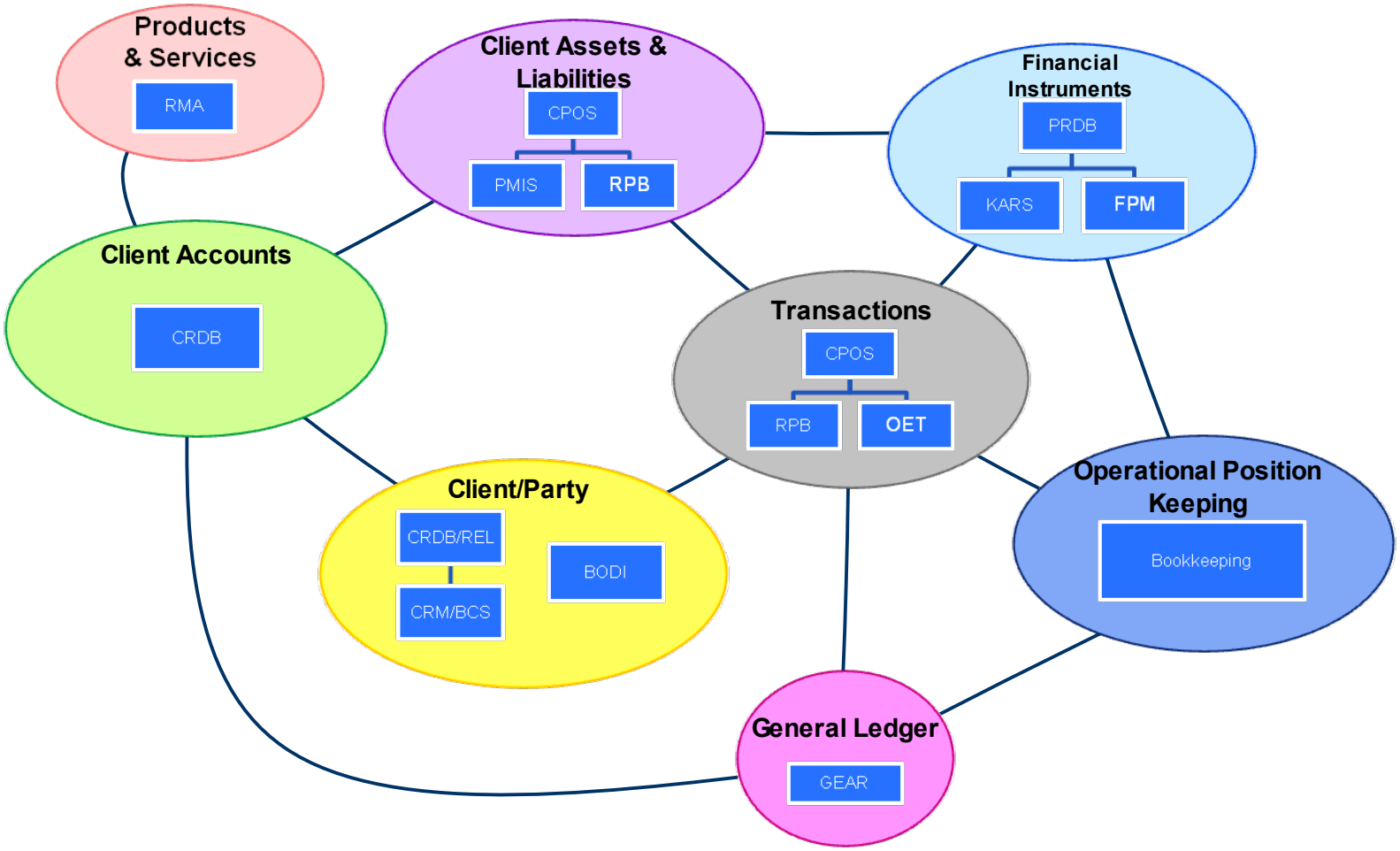
Target State – Data View Arranged by Subject Areas

As the data level, Applications and data stores will be arranged into Subject Areas – Data Services will serve up each Subject Area - Each Subject Area will be presented to applications and reporting via EDM to ensure consistent view and single source while allowing applications to optimize processing.



Top of hierarchy indicates master - Access and Security System not represented

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The Target Data Architecture supports data needs across applications

As the enterprise features of the data architecture are implemented, application-specific databases will be decommissioned and we propose a POC that will hopefully lead to the requirement serviced directly from the source through a virtualization layer.

Any number of delivery and presentation channels can be built on top

A set of well defined, supported, and governed tools, and only that set, provides reporting

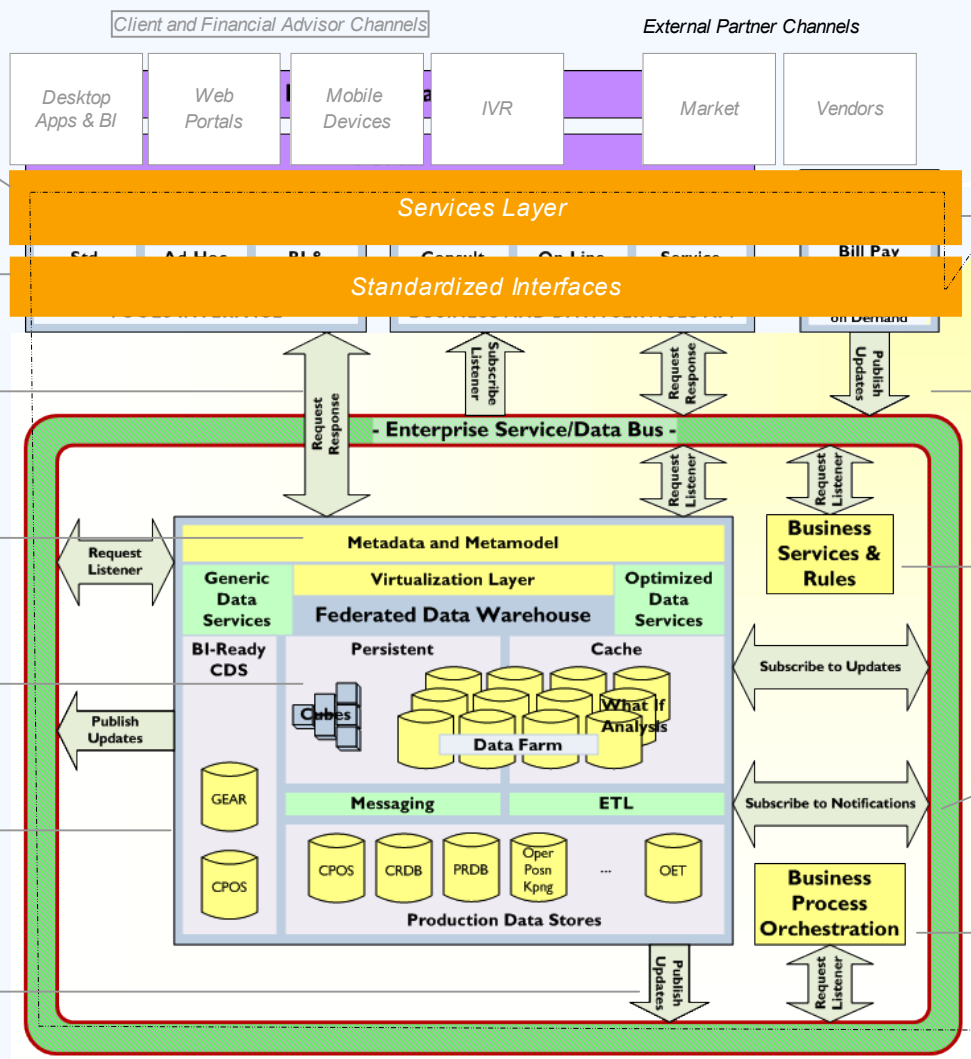
All data access utilizes the same underlying data model and services

BI Conformed Data Store virtually accessible through Metadata layer

Dynamic configuration of core cache and distribution increases agility

Existing systems that meet all requirements continue to serve their own analytical data

Data Update notifications are available for immediate processing



All Applications use the same API and underlying services to access data

Interaction with external data vendors through common services

Business calculations and rules available to applications part of data services

As real-time data becomes available, it is stored, transformed, and routed to applications that need it

BPO is separated from core data processing for common use

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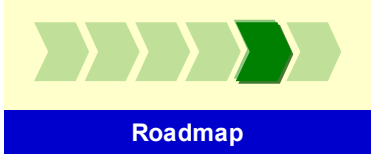
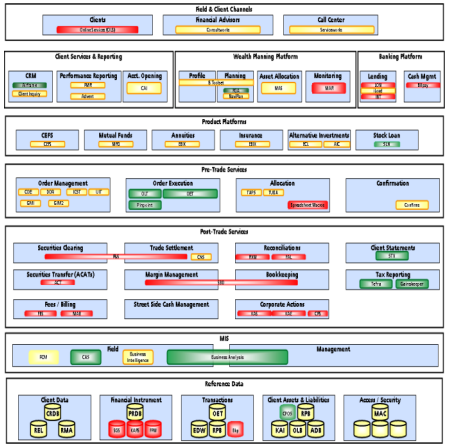
Transformation Roadmap

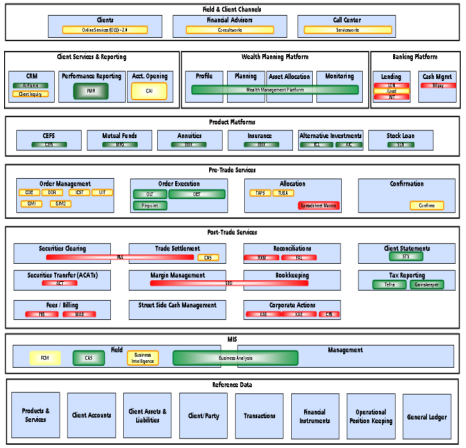


Current State



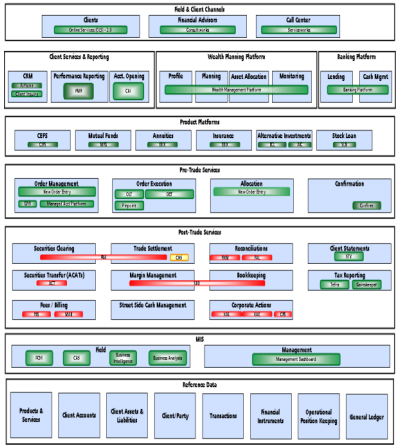
- The current state is not fit for purpose, is overly complex and highly fragmented.
  This complexity impedes the business rather than enables it
- IT Architecture Target Target-State is developed as to serve as the core foundation of the investment governance process/decision making

2011 Focus



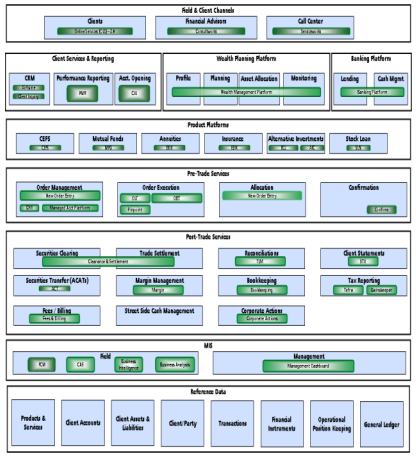
- Year 1 – in 2011, we will naturally must focus on enhancing the platforms that will support revenue increase (e.g. Insurance & Annuities, Structure Products, AI, etc) and improve FA and Client experience with front-to-back workflows
- In addition, it is critical that we effect the strategic core architecture changes we need to reach the target state (TS) e.g.:
  - Consolidate data stores onto golden sources
  - Re-design front systems that decouple presentation layer from business logic
  - Deployment of key components of messaging infrastructure as we re-platform and refresh the application portfolio
- Leverage Group technology for re-use opportunities and facilitate infrastructure demand planning with GTIS

…



- Year 2 - 3 –Progress is measured by degree of implementation of the Target-State architecture tenets
- Tactical projects are made faster, cheaper to execute as the Target-State architecture components are deployed

Target State



- Strategic roadmap is in in place for all major area of the Target-State model
- Increased pace of system rationalization as several multi-year programs reach targets
- We will continue to invest in areas with completed strategies in support of business demand

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Implementation Activities (1 of 2)

Domain	Name	Project / Project Objective	Visible Benefit
Data Lifecycle Management	01-01 - Establish Data Service Framework	Develop Data Service Framework to be used by all golden sources (Eliminate SGS)	Elimination of Point to Point Interfaces; Easier integration in WMA Environment
	01-02 - Financial Instrument Master	Enhance PRDB in order to enable the elimination of redundant data stores (eliminate SGS)	Single source for financial instrument data
	01-03 - Client Master	Enhance CRDB to be WMA's single client master responsible for all client and prospective client reference data (e.g. relationships, etc)	Migrate to client-focussed model away from account centric paradigm.
	01-04 - Transaction Master	Develop OET database to be the central orders, executions, and trades data store	Single source for all central orders, executions, and trades that impact an account
	01-05 - Entitlements Master	Consolidate multiple entitlement data stores across the Front Office platform onto a single data store	Enhanced security and entitlement posture; Single control point for entitlement information
	01-06 - Client Transaction, Asset & Liabilities and Transaction Master	Master all client positions, transactions including all assets and liabilities for client data	Full view of client balance sheet and transactions
	01-07 - Client Document Master	Single repository for client documents (e.g. Templates, Completed Forms, Communications)	Full view of correspondence with a client
	01-08 - Data Governance	Implement Data Governance process in partnership with the Business owners of the data	Consistent data definitions that can be referenced throughout the organization
	01-09 - Virtual Data Warehouse	Virtual federated data warehouse with common metadata layer, enable enhanced analytics	Enables the consolidation of multiple reporting data stores to enable analytics without the overhead of a physical data warehouse
	01-10 - Improve Business Intelligence Platform	Ad-hoc data reporting and data mining / analytics (possibly leveraging new data abstraction layer)	Agile data mining, instant analytics
Application Remediation (Data)	02-01 - Convert to Universal Account #	Remediate Applications to Remove business logic from legacy Account number, move onto universal account number	Less prone to data privacy issues (e.g. unautorized access to accounts)
	02-02 - All Client Data read from CRDB	Remediate applications to read CRDB for client data using the common service	Consistent client data used end to end
	02-03 - All Financial Instrument Data read from PRDB	Remediate applications to read PRDB for financial instrument data using the common service	Consistent financial instrument data used end to end
	02-04 - All Client Transaction Asset & Liability Data read from CPOS	Remediate applications to read CPOS for client position data using the common service	Consistent client position and transaction data presented end to end
	02-05 - All Transaction Data read from OET	Consildation of orders, executions, and trades data store in OET	
Post Trade Services	03-01 - Real-Time Positions and Balances	Enhance RPB to capture cash balances and consolidate "RPB" like data stores onto single instance (Might be throw away if outsourcing scenario)	Enables Banking Strategy; Real time view of client position and balance
	03-02 - Remove Margin from Bookkeeping	New Margin Calculation module that isn't embedded with Bookkeeping system and replatform Margin Interest calculation engine (Might be throw away if outsourcing scenario)	Start de-risk bookkeeping platform by de-coupling Margin
	03-03 - Replace Core Platform	Replace core functions of the backoffice: Bookkeeping, P&S, Reconciliations, Margin, ACATS, Corporate Actions, Clearing and Settlement	New backoffice platform that can support all products without manual intervention; De-risk backoffice
	03-04 - Front to Back Workflow Engine	Deploy a front to back business process workflow infrastructure (in conjunction completing the build out of the Servicework platform)	Elimination of Manual Processes
	03-05 - Convert to Group Reconciliation Utility	Migrate reconciliations onto TLM with market information	Consistent with Group Tech Startegy; Enable Group Reconciliation Utility
	03-06 - Consolidation of Fee Billing Applications	Complete the build out of single fee billing application and decommission legacy applications	Enable different view across client and assest. Provide transparency of revenue structure from accurate billing of fees across products

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Implementation Activities (2 of 2)

Domain	Name	Project / Project Objective	Visible Benefit
Client & FA Experience	04-01 - Rewrite Client Frontends	Re-write OLS under a consistent mobile-enabling framework utilizing roles and entitlements to deliver functionality rather point applications	Consistent look an feel; Improved usability and mobility
	04-02 - Rewrite FA Frontends	Re-write Consultworks platforms under a (mobile-enabled) consistent framework utilizing roles and entitlements to deliver functionality rather point applications	Consistent look an feel; Improved usability and mobility
	04-03 - Source all Client Reports from Gold Sources	All front office systems should pull client position and other data from Golden Sources	Consistent view of the client across front-end platforms
	04-04 - Enhanced Account Opening Process	Enhance the account opening process to incoporate Document Management strategy	Enhance client experience
	04-05 - Client Entitlement and Role-Base services	Role based access through a common interface to provide consisten access and entitlements based upon role	
	04-06 - Improved Performance Reporting Aligned with Client Goals	Complete PMR enhancements in order to decommission Axys Advent	Integrate performance reporting based on CPOS data
	04-07 - Usability, Touch and Feel	Improve the look and feel of Consultworks and OLS	Enhance Client & FA Experience
	04-08 - Integrated Contact Management Application	Convert remainin FA population onto new CRM system. Integrate CRM application into telephony and Consultworks platform	Industry leading CRM capability
	04-09 - Practice Management Tools	Restructure dashboard and management tool to reflect emphasis on FA benchmarking	Implem,ent new businbess model and strategy with focus on customert goals
	04-10 - Computer Telephony Integration	Integrate (IP) telephony dimension to customer interaction model	"Push to Talk", Integrated Communication, Campain Management, Improved Client Experience
Order Entry & Trading	05-01 - Consolidate Order Entry Applications	Single Order Entry front end that routes to the appropriate order management application. Consolidate order management applications from 12 to 6	Simplified Order Entry process for the FA; Rationalized application suite
	05-03 - New Managed Acct Trading Platform	Enhance or replace ICST	True Managed Account Trading Platform that enables true Asset Allocation
	05-03 - IB Integration	Leverage IB trading infrastructure (e.g. Oscar, Pinpoint)	Lower cost of ownership; Leverage Group Infrastructure
Wealth Management	06-01 - One Client Profile	Move towards a w holistic client profile capture methodology and view of assets and liabilities	Full view of a clients portfolio that includes external accounts
	06-02 - Integrated Wealth Management Platform	Consolidation of investment tools under an integrated framework that allows for internal and external sourcing of capabilities	Integrate Wealth Planning that enables the FA to ubetter evaluate risks and goals (less product centric)
Banking	07-01 - New Cards Platform	Insource existing credit card platform	Improved Credit Card experience
	07-02 - Enhance Mortgage Platform	Upgrade mortage platform to support Business desire to scale the mortgage business	Enable Mortgage Startegy and Integration
	07-03 - Upgrade Electronic Billpay	Improve online bill pay experience	Improved Client Experience
Products	08-01 - Alternative Investment Automation	Automate AI order entry process to enable straight through processing and scalability	Elimination of manual processes required to support booking of AI investments
	08-02 - Enhance Annuities & Insurance Platform	Insurance and Annuity platform	Able to scale Annuities and Insurance business by eliminating manual process
	08-03 - Mutual Fund Back office	New reconciliation tool for Mutual Funds	Enable automated settlement of Mutual Funds
	08-04 - Stock Loan		Improve utilization of WMA box
	08-05 - Structured Products		
	08-06 - CEFS	Only upgrade current CEFS platform to meet required commitments to high revenue clients. Focus on consolidation onto WM&SB platform	Reduce cost of mainteing two platforms and compliance with regulatory mandates
Supporting Infrastructure Changes	09-01 - OneTransactionKey	Universal transaction ID; consistent immutable internal IDs for any business entity (transaction, client, loan, account)	Traceability
	09-02 - Common Error Handling		

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Summary of Benefits , Success Factors and Implementation Considerations

Summary of Benefits	Success Factors	Implementation Considerations
• Customer-centric view of the business • Faster time-to market performance • Higher productivity of FAs • Enhanced end-user experience • Better understanding of risks and exposures • Reduced manual do-over • Reduced operational risks • Lower costs with better CTB/KTLO cost ratio • Lowered barrier of entry in new business • Faster product development	• Collective understanding and agreement on principles, tenets and relative priorities • Validate plan with all constituents Systematic planning with attention paid to dependencies within and outside of WTA • Validate/Insure that next year and long term investments are aligned with strategy • Structured change management infrastructure including: – Efficient governance model including business and CTO components – Mobilization of change agents – KPIs/Balance scorecard consistent w ith strategy – Communication plan geared at each constituencies	• Workshop session within next 3/4 weeks • Consider elements of Design for Six Sigma for implementation planning and execution: – Voice of the Business, Voice of the developers etc… – Tollgates – Governance • Consider real option pricing in developing business case for investments • Plan execution of task to support the deployment of new architecture tenet (As opposed to having massive cross-organizational initiatives) • Develop/Use scenarios to stress test proposed plan • Make this a showcase on how the CTO function can support change

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The current definition of our proposed initiatives need to be re-evaluated and probably re-scoped

As we begin to formalize the scope of each of these programs, IT and the Business need to work together to ensure that programs aren't only concerned with specific subset but rather help move the dial towards the target architecture.

Measures to what degree each initiatives (Column) can be scoped to support movings towards the target architecture (Tenets - Lines)

	Line Total (Normalized) Measures how much the Initiatives will make the Tenets happen	Mandatory										Banking							Marketing					COO					WMAG				Solutions	
		Regulatory / Mandatory	Safety & Soundness	CPOS	Performance Reporting	ARS	Global Risk	Finance	IT Continous Improvement(e.g. Data Centre, CMOD)	Discretionary Bucket	Reserve	Mortgages	Cards	SBL	Billpay	Banking	Pricing	FA Branding	Data Market	Client Experience	WMR	Workflow	Operations	Client Experience (NAO / Forms)	Realtime Position & Balances	Margin	FA Comp	Puerto Rico	Productivity	Private Bank	CEFS	Pricing
Complete WMA Enterprise Data Model -- Next Phase of WMA Data Compass standardizing entity mapping, data dictionaries, data quality and management practices, including data governance	02																																	
Multi Channel -- Extension of services framework, targeting multi-channel consistency for customers and FAs and CRM	02																																	
Converged Common Business Workflow platform -- eg Build out TBCO Process frameworks and eliminate "hard-coded" business process logic from applications to enable STP, e.g. Account Opening TX	02																																	
Standardized System Interfaces With Reusable Services Mature the services catalog across distributed and mainframe systems as well as internal (e.g. B) and external (outsourcing) service vendors, retire point-to-point interfaces over time	05																																	
Single Client Master for Client Positions and Client Data -- Distributed master client data and catalogs accounts, hierarchies / house holding. Master all client positions including trading, banking, loan balances	04																																	
Standardized Reporting Tools -- Mature the reporting platform to homogenous reporting stack, remove "hard-coded" business logic and calculations to make them shareable, and develop a self service reporting solution	02																																	
Standardized common messaging framework across distributed and mainframe applications -- Leverage TBCO infrastructure for both distributed and mainframe systems	03																																	
Client- and Process-driven document management -- Migrate to centralized document management and common document generation platform, e.g., e-forms and proposals	01																																	
Converged Content, Collaboration, Portal Solutions -- Simplify FA & client experience through common platform for content generation and management, document management, and collaboration	01																																	
Rationalized Web Portals -- Consolidate Client-facing interfaces including CEFS and OLS	02																																	
Enabled Alternate Sourcing -- Lay the functional services and data framework to provide flexibility in process outsourcing	02																																	
Integrated Records Management Map - WMA document catalog and assure records retention compliance	01																																	
Componentized Aggregation Framework based on Group Functions -- Model separate functions and rationalize redundant applications	02																																	
Single real-time, data sourcing and distribution channel -- Develop Data Compass for golden sources and common data services layer for distribution of WMA reference and operational	03																																	
Market Data Utility -- Develop a Market Data Strategy for sourcing and distributing market data to WMA applications with possible sharing with other Bus	02																																	
show each initiative moves us to the Target Architecture		012	035	051	026	000	000	005	021	000	000	044	008	000	023	048	006	026	000	026	011	082	027	083	043	043	024	000	020	000	011	000		

Our current plans are generally formulated in an application-centric way (application development, re-platforming…) (e.g. CEFS)

… or important but tactical constraints (Puerto Rico) …

… and although some planned investments will de facto help achieving the target state (Developing a single client master for client position and data as part of the CPOS initiative)…

… investments are either too broadly (Data Market?) or too narrowly defined (Insurance?)

… and some critical initiatives are not explicitly covered

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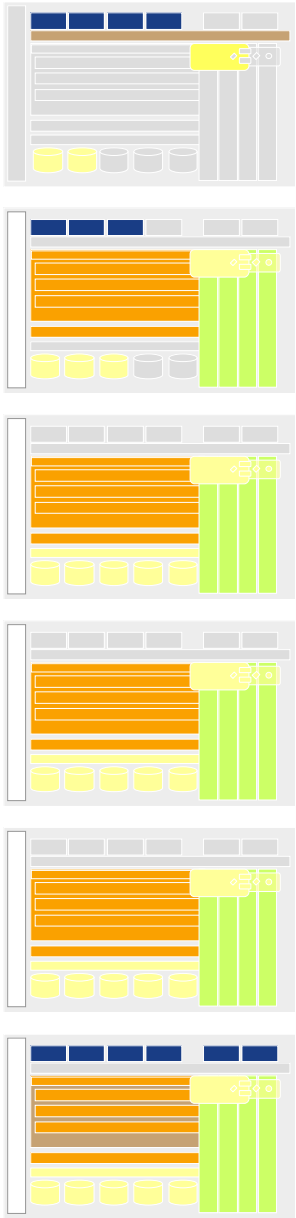
Case Study: Supporting Activities for Wealth Planning Solutions and How They Will Benefit the Overall Implementation of the Target-State Architecture

Roadmap

Design and implementation within core architecture principles

Domain	Name	Objectives	Benefits
Wealth Planning	06-01 - One Client Profile	Move towards a holistic client profile capture methodology and view of assets and liabilities	Full view of a clients portfolio that includes external accounts
	06-02 - Integrated Wealth Management Platform	Consolidation of investment tools under an integrated framework that allows for internal and external sourcing of capabilities	Integrate Wealth Planning that enables the FA to better evaluate risks and goals (less product centric)
Products	08-01 - Alternative Investment Automation	Automate AI order entry process to enable straight through processing and scalability	Elimination of manual processes required to support booking of AI investments
	08-02 - Enhance Annuities & Insurance Platform	New Insurance and Annuity platform	Able to scale Annuities and Insurance business by eliminating manual process
	08-03 - Mutual Fund Back office	New reconciliation tool for Mutual Funds	Enable automated settlement of Mutual Funds
	08-04 - Stock Loan		Improve utilization of WMA box
	08-05 - Structured Products		



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We also need to develop KPIs that can measure how we implement the key tenets of the target IT Architecture

This is a pro-forma model to be refined and discussed with all shareholders

Key Tenets	Target State Key Performance Indicators
Investment	% of initiatives budget earmarked with strategically-aligned initiatives
Complete WMA Enterprise Data Model	Drive participation in IT and Business Data Governance process and completion of enterprise data model (currently 35%)
Client- and Process-driven document Mgmt	XX % migration to WMA document management; target 100%
Single Client Master for Client Positions and Client Data	Eliminate data stores which are redundant with CPOS, RPB, and CRDB in 2011-12 % adoption of compliance with golden sources
Integrated Records Management Map	XX % migration to WMA document management - Measure compliance with Group policies
Enabled Mobile/Alternative Channels	Set dates for business and date services availability and adoption for front-end access
Standardized Reporting Tools	% migration to standardized reporting tools while externalizing business logic to shareable business services layer
Enabled Alternate Sourcing	Define and adopt external services interfaces and common data model for vendor interactions
Converged Common Business Workflow platform	Target 1MM cases monthly through ServiceWorks, increasing to XX by YE 2011 14 iProcess and BusinessWorks apps increasing to __ by 2011
Componentized Application Framework	Application rationalization targets 46% reduction of key applications in Target state (from 241 to 112. 50% reduction of data stores from 24 to 12 XX% of apps reviewed and design plan for loosely coupled component approach
Standardized common messaging framework across distributed and mainframe applications	Drive adoption of data services layer to data consuming applications; 100% of apps should have adoption date planned
Rationalized Web Portals	Adoption dates set for client-facing and FA-facing framework, set timeline for mobile services
Market Data Utility	Set adoption date for Group-wide solution
Single, real-time, data sourcing and distribution channel	Set implementation roadmap of vertically defined front-to-back data distribution
Standardized common messaging framework across distributed and mainframe applications	2011 focus on assessment and development of new services within standard protocols and remediation planning XX % of transaction and interface protocols have adoption dates planned
Converged Content, Collaboration, Portal Solutions	.X% application remediation of hard-coded business logic; go-forward, all new applications to expose business logic layer

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Next Steps

- Validation and syndication of existing proposal for tenets, portfolio of initiatives and pro-forma plan:
  - Executive level - WMA and Group IT
  - Other stakeholders - WMA and Group IT
- Next steps within 2/3 weeks:
  - Syndication of IT Architecture Strategy
  - Prepare and contribute to workshop set up to develop front to back strategy and investment plans for WMA (in progress)
- Set up / strengthen infrastructure for implementation and governance management
  - Agree on success factors (Including communication and training)
  - Build up scorecard based on agreed upon KPIs
  - Understand linkage with individual objectives
  - Link with existing PMOs
  - “Air Traffic Control” model
- Next level of planning
  - Finer-grained dependency analysis within current plan
  - Other views/ dependencies, e.g. Security, Infrastructure
  - Program planning proper

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F Managing the Transformation: Integrated Timeline, Investments and KPIs

G Next Steps

A – Participants

B - Using Design for Six Sigma tools for Architecture Strategy Definition

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Attachment A - Participants

Executive Summary	Bella Allaire Robert Anselmo Jose Alayo Gary Blumenthal	Vincent Campagnoli Stu Feld Padmini Chittipeddi	David Lesser Scott Mazur John McKeon Mike Oliveira	David Small Minerva Tantoco Manuel Barbero	Jonathan Springer Jan Stobbe Sunil Jadhav	Russ Le Blang James Nordstrom Zoltan Majdik
Wave 2	***Banking***		***Investment Solutions***			
	Robert Anselmo Mike Memoli Jonathan Springer		Mike Oliveira Darren Greco Sridhar Ganapathy	Kimanshu Shah Durga Koganti	Kaushik Mehta Jereme Conti John Rebimbas	Henry Villafane Leonid Grinberg Jan Stobbe
Wave 1	***Post Trade***		***Order Entry and Trading***		***Front Office***	
	Tony Porraspita Dmitry Goncharuk Arlene Richman Andre Kupelian Lana Feld David Small Alan Scudi	Ella Gleizer Henry Villafane Tom Laba Mark Burke Tony Barone Tim Wald Jan Stobbe	John Mckeon John Rebimbas Tony Porraspita Erick Chang Dmitry Goncharuk	Mike Oliveira Igor Fuksman Teo Demetriades Jan Stobbe Sunil Jadhav	Alex Beigelman Padmini Chittipeddi Mike Edelstein Jonathan Springer	
					Client Information	
					Alex Beigelman Vlad Leshinsky Jonathan Springer	
Data Current & Target States	Sunil Jadhav Jonathan Springer Sam Moeller Vladimir Leshinsky David Hunter Abraham Schwartz	Jaime Vallecilla David Lesser Andre Kupelian Melanie Dorn Rachel Weiss	Tom Grande Andrew Depasquale Stephen Cook Henderson Cole Tony Barone Robert Anselmo Ronnie Wang	Sergey Zaster Ryan E. Long Max Loshinsky John Francis Tony Porraspita Alla Turetsky	David Small John Rebimbas A. Stankevitch Ronnie Wang Olga Polishuk Scott Leavitt	Matt Clayton Hong Jiang Agustin Gonzalez Ella Gleizer Teo Demetriades David Porter Ed McGee

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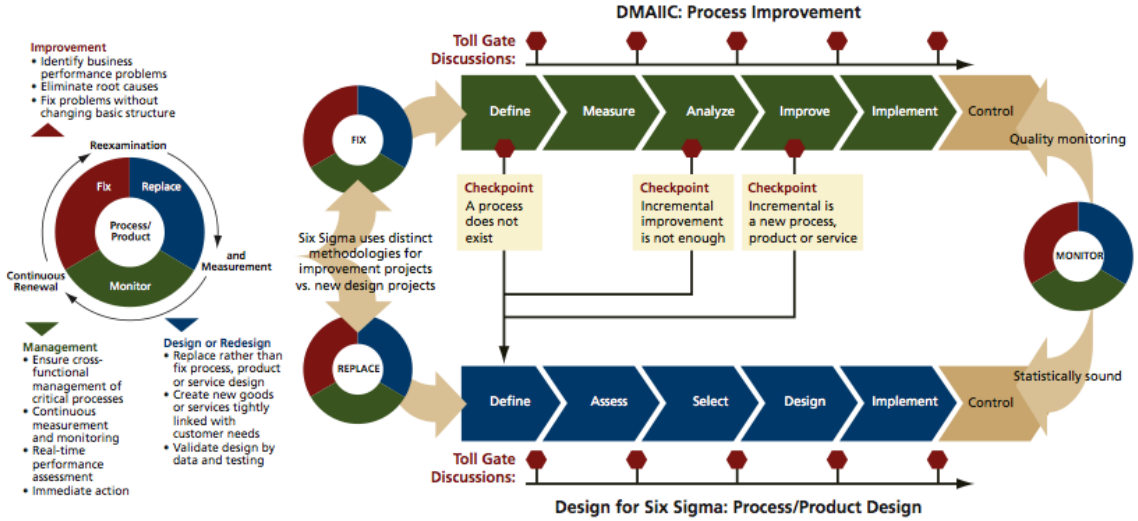
Attachment B - Using elements of the Design for Six-Sigma toolkit for IT Architecture Strategy development and Activities prioritization

Design for Six Sigma (DFSS)

- The Six Sigma concepts and toolset are often used for the improving or managing ***existing*** process, often using statistical analysis and other data-oriented techniques. This is usually accomplished using a combination of key steps — Define, Measure, Analyze, Improve, Implement and Control — referred to collectively as the **DMAIIC** steps.



- There is, however, another path through the Six Sigma world, aimed at defining ***new*** processes, an approach that can be used ***if a process does not even exist***, if ***incremental improvement is not enough*** or if what is needed is an entirely new process, product, service or solution.
- In essence DFSS is ideally suited to deal with issues that are ***neither trivial nor necessarily well defined*** and/or when the ***issues at stake are important and cut across organizational boundaries***, which is typically true for IT Architecture-related projects.
- In the Design for Six Sigma (DFSS) framework, defining an entirely new set of processes requires a new approach, based on the execution of a new set of different action steps — Define, Assess, Select, Design, Implement and Control.

This attachment does not attempt to further describe the DFSS approach to problem solving and focus instead on the DFSS tools that were used in the definition of the proposed End State Architecture Strategy development in this presentation. It is however expected that the use of other DFSS construct (e.g. the generalized used of the Voice of xxx process, the development of a strong "Define" phase at the onset of the project, the use of well-defined tollgates and the DFSS approach to project governance and the definition of KPIs aligned with the QFDs) will greatly improve the execution of the proposed strategy.

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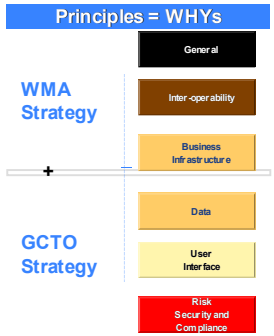
How we used DFSS tools (Pair-Wide Comparisons and Quality Function Deployment) in developing the proposed strategy

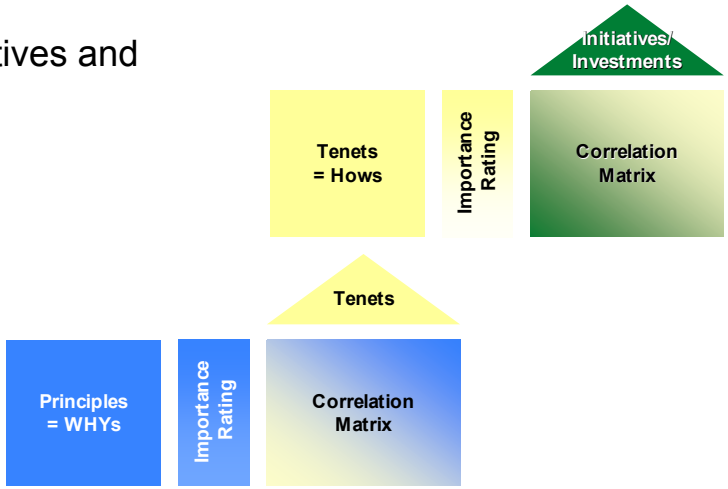
- Pair-wise comparison is a simple yet powerful way of quantifying the relative importance of options within a given list, it ranks the “WHYs”:
  - In this document the set of technology Principles are the drivers of the IT Architecture Strategy, “WHY” we are developing it -- They are expressed in terms relevant to the IT community members in a way that translate their understanding of the business AND their professional judgment as to what these principles should be.
  - This said, not all principles carry the same weight and it is useful to rank these principles so that the recommendations in the strategy reflect this ranking.



Description	Normalized Relative Importance Score
Define Unique Golden Sources & Assign Business and IT Ownership for each Data Object	0.59
Enable Consistent Client and FA / CSA / Operations View Sourcing the same Data	0.57
Avoid Business Logic in Front-end Components	0.56
Provide Channel-independent Solutions (Multi Channeling, eg Web N/R, Mobile, Contact Center)	0.56
Reduce Redundant Business Data Stores Allow Controlled Data Redundancy, under Control of Data Owner	0.55
Enable Flexible and Efficient Product Configuration (Order Mgmt Framework)	0.54
Provide User Group Specific Platforms Via Common Framework (e.g. FA, Ops, Client)	0.53
Follow Core Data Quality Principles (eg Correct at Source)	0.50
Use Standard Communication and Interaction Patterns between Applications (eg event-based)	0.46
Design for alternate sourcing	0.46
Strive for Self-contained Loosely Coupled Applications	0.45
Follow the Group Functional Model (CBM)	0.44
Provide Multi Currency Capability	0.42
Re-use Software Components / Services / Solutions	0.42
Design for Failover and Recoverability	0.42
Strive for cost-efficient design and development	0.41
Use Standard Systems and Protocols for External Communication	0.40
Standardize Business Infrastructure such as Doc Mgmt, Msg, Workflow)	0.38
Support an Evolutionary Approach	0.35
Architect different Systems for Operational, Analytical and Archiving Purposes	0.25

- Typically, it is difficult to order a full list of principles. It is however much easier for anyone to express an opinion as to how Principle “A” rates vis a vis Principle “B “, e.g. “Much more Important”, “More Important”, “A bit More Important”, “As Important”, “A Bit Less Important”, “Less Important”, “Much Less Important”.
- A pair-wise comparison exercise consists in filling up a matrix where the top line and the left-most column are the items to sort and then assigning a numerical rating that express the relative importance of the items.

- QFD is a general approach that helps both prioritize objectives and trace how initiatives and investments should be prioritized
  - QFD analysis enables transformations from Drivers (“WHYs”) to the key Tenets of the architecture, i.e. HOW any activities or initiatives/investments should be designed to ensures traceability to the initial Principles.
  - The same process can then performed between the tenets and the proposed initiatives/investment.
  - By introducing traceability from investments and activities to ranked principles, QFD really helps technology strategists focus on what matters.



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Pair-wise Comparison Implementation

WMA – QFD Analysis – 2010-10-11B.xls

	Choose from the pull-down menu in the yellow cells - Do not modify greyed cells. The rating should read "Item on TOP matters (More/.../Less) than item on LEFT"	Use Standard Communication and Interaction Patterns between Applications (eg event-based)	Strive for Self-contained Loosely Coupled Applications	Provide Multi Currency Capability	Standardize Business Infrastructure such as Doc Mgmt. Msg, Workflow)	Re-use Software Components / Services / Solutions	Enable Flexible and Efficient Product Configurati (Order Mgmt Framework)	Architect different Sy Analytical	Defi	nal Model (CBM)	port an Evolutionary Approach	Strive for cost-efficient design and development	Design for Failover and Recoverability	Design for alternate sourcing	TOTAL	Normalized rate (0 > 1)
Interoperability	Use Standard Systems and Protocols for External Communication	... (3) matters more than ...	... (1) matters just a wee bit more than ...	... (9) matters much more than ...	... (3) matters more than ...	... (3) matters more than ...	... (1) matters just a wee bit more than ...	... (-3) matters less than ...	... (9) matters mu more than ...	... (0) matters as much as ...	... (3) matters more than ...	... (0) matters as much as ...	-45	0.40
Interoperability	Use Standard Communication and Interaction Patterns between Applications (eg event-based)		... (0) matters as much as ...	... (-1) matters just a wee bit less than...	... (-1) matters just a wee bit less than...	... (0) matters as much as ...	... (3) matters more than ...	... (-3) matters less than ...	... (3) matters than ...	... (-3) matters less than ...	... (3) matters more than ...	... (0) matters as much as ...	-18	0.46
Interoperability	Strive for Self-contained Loosely Coupled Applications			... (-1) matters just a wee bit less than...	... (-1) matters just a wee bit less than...	... (1) matters just a wee bit more than ...	... (3) matters more than ...	... (-9) matters much less than ...	... (3) mat tha	... (-3) matters less than ...	... (3) matters more than ...	... (0) matters as much as ...	-22	0.45
Business Infrastructure	Provide Multi Currency Capability				... (3) matters more than ...	... (3) matters more than ...	... (3) matters more than ...	... (0) matters as much as ...	... (3) matte than ...	... (3) matters more than ...	... (3) matters more than ...	... (-1) matters just a wee bit less than...	-33	0.42
Business Infrastructure	Standardize Business Infrastructure such as Doc Mgmt. Msg, Workflow)					... (-1) matters just a wee bit less than...	... (3) matters more than ...	... (0) matters as much as ...	... (9) matters mu more than ...	(0) matters as much as ...	... (3) matters more than ...	... (3) matters more than ...	-54	0.38
Business Infrastructure	Re-use Software Components / Services / Solutions						... (3) matters more than ...	... (-3) matters less than ...	... (3) matters mo than ...	. (0) matters as much as ...	... (3) matters more than ...	... (0) matters as much as ...	-34	0.42
Business Infrastructure	Enable Flexible and Efficient Product Configuration (Order Mgmt Framework)							... (-9) matters much less than ...	... (1) matters j wee bit more ...	... (-3) matters less than ...	... (3) matters more than ...	... (0) matters as much as ...	18	0.54
	Architect different													

- The items to be sorted (Principles) are arranged in the top line and the left-most column
- At cell at line i, and column j, a rating of (9,3,1,0,-1,-3,-9) is assigned based on the relative importance of the item in the line ii in the first column vis-a-vis of item on column j in the first line of the table
- For instance as a principle, the “Re-use of Software Components / Services / Solutions” was deemed just a bit less than important than “Standardiz(ing) Business Infrastructure such as Doc Mgmt. Msg, Workflow)”
- … but using standards communication and interaction patterns between applications (e.g. event-driven patterns) was deemed more important (Rating = 3) than using Standard Systems and Protocols for External Communication
- An inverse weight is automatically assigned to cell j,i and rating are added up across the table to provide a relative rating that capture the overall importance of the principle against others
- This rating is then normalized (the maximum range is from -9 to 9 times the number of columns to provide the ratings on page 7

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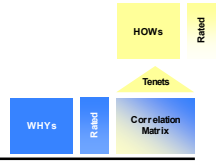
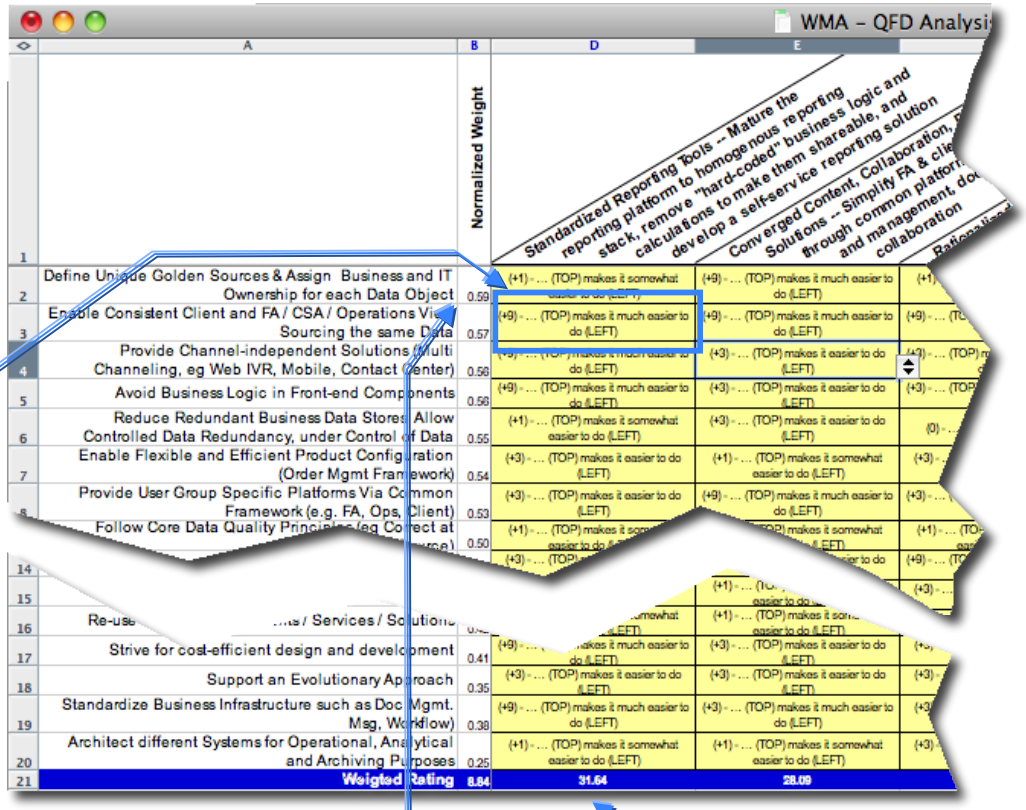
QFD Implementation Example: Linking Tenets and Principles



- The key Principles are reported on the first column (0) of the table (WHYs);
- A series of key Tenets (HOWs) were identified in the deep-dive sessions. They listed on the first line of the table;
- On a cell i, j we evaluate to what extent a proposed architectural Tenet on column j will support the principle on line i;
- For instance, it was estimated that "Using a standard service stack to support reporting functions…" will make it much easier to provide "A consistent view to users… sourcing the same data (**rating = 9**)
- We then evaluate scores by column by totaling the weighted correlation for each Principles and Tenets
- For example:
  - The relative importance of the Principle "Define Unique Golden Sources & Assign Business and IT Ownership for each Data Object" as defined by the pair-wise comparison process is **0.59**
  - Therefore the score "9" in the cell mentioned above contributes 9 x 0.59 = 5.31 to the overall total (**31.64**) scored by the tenet "Standardized Reporting Tools -- Mature the reporting platform to homogenous reporting stack…"
- **The results (bottom scores) provides an objective way of prioritizing the "HOWs" / Tenets (aka Critical To Quality (CTQ) factors in DFSS parlance) – Presented on Page 9**